Commitment and Randomization in Communication

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Prosecutor-Judge Example

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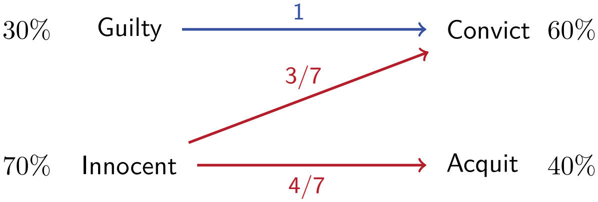
Judge's preferences: convicts if the probability of guilt $\geq \frac{1}{2}$

Prosecutor's preference: always prefers conviction.

Prosecutor-Judge Example

Optimal Information Structure



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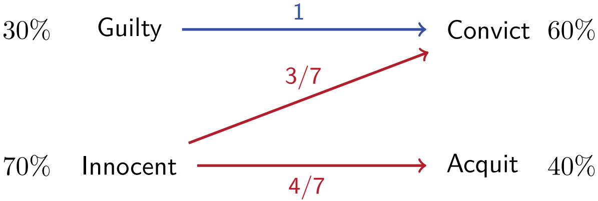

Value of Commitment: $V_{\text{Bayesian Persuasion}} > V_{\text{Cheap Talk}}$

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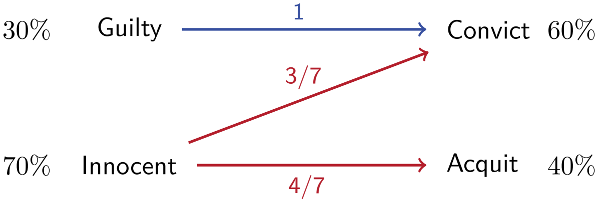

Value of Commitment: $V_{\text{Bayesian Persuasion}} > V_{\text{Cheap Talk}}$

Value of Randomization: $V_{\text{Bayesian Persuasion}} > V_{\text{BP w.o. randomization}}$

Theorem 1. In finite generic persuasion models, a Sender values commitment if and only if she values randomization.

Bayesian Persuasion

Cheap Talk

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$\Delta_{\text{commitment}}$

Cheap Talk

$$\begin{tikzcd}[column sep=huge, row sep=huge]
\text{Bayesian Persuasion} \arrow[r, dash, "\Delta^{B}_{\text{randomization}}"] \arrow[d, dash, "\Delta_{\text{commitment}}"'] & \text{Persuasion without Randomization} \\
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Theorem 3.

- The share of preferences such that commitment has no value is at least $\frac{1}{|A|^{|A|}}$.
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E.g., when $|A| = 2$, commitment has no value in at least 25% of the preferences.

Literature

Under Transparent Motives:

- Sender obtains ideal payoff through cheap talk or Sender values commitment (Lipnowski-Ravid '20)
- Sender obtains ideal payoff at prior or Sender values randomization (Best-Quigley '24)
- Cheap talk = Persuasion $\Leftrightarrow$ Mediation = Persuasion (Corrao-Dai '24)
- Verifiable messages = Persuasion $\Rightarrow$ Randomization has no value (Titova-Zhang '25)

Commitment and Randomization by Receiver in Disclosure Games:

- Glazer-Rubinstein '06; Sher '11; Hart-Kremer-Perry '17

Model

One Sender (S) and one Receiver (R)

State: $\theta \in \Theta$ with prior μ_0

Action: $a \in A$

Message: $m \in M$

All spaces are finite

Payoff functions: $u_S(\theta, a)$ and $u_R(\theta, a)$, take values in $[0, 1]$.

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An environment: $(u_S, u_R) \in [0,1]^{2|\Theta||A|}$

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A claim holds for x-share of environments if the set of environments where the claim holds has Lebesgue measure x.

Strategies

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Receiver chooses an action strategy $\sigma_R : M \to \Delta(A)$.

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A profile of strategies (σ_S, σ_R) induces expected payoffs for $i = S, R$:

$$U_i(\sigma_S, \sigma_R) = \sum_{\theta,m,a} \mu_0(\theta)\sigma_S(m|\theta)\sigma_R(a|m)\, u_i(\theta, a)$$

Sender-IC and Receiver-IC

A profile (σ_S^*, σ_R^*) is Sender Incentive Compatible (S-IC) if

$$\sigma_S^* \in \underset{\sigma_S : \Theta \to \Delta M}{\arg\max} \; U_S(\sigma_S, \sigma_R^*)$$

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A profile (σ_S^*, σ_R^*) is Receiver Incentive Compatible (R-IC) if

$$\sigma_R^* \in \arg\max_{\sigma_R : M \to \Delta A} U_R(\sigma_S^*, \sigma_R)$$

Cheap-talk equilibrium is a profile that satisfies (S-IC) and (R-IC).

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Partitional persuasion profile is a profile that satisfies (R-IC) and σ_S is a pure strategy.

- **Partition payoff**: the maximum U_S induced by a partitional persuasion profile.

Theorem 1. For a generic set of preferences,

Persuasion Payoff $=$ Cheap-Talk Payoff

if and only if

Persuasion Payoff $=$ Partition Payoff

Proof of “If”

WTS: if a partitional persuasion profile (σ_S^*, σ_R^*) achieves the persuasion payoff, it is also a cheap-talk equilibrium.

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So both σ_S^* and σ_R^* are pure strategies.

$$
\begin{tikzcd}
\cdots \arrow[dr] & & \\
& m \arrow[r] & a \\
\theta \arrow[ur] & &
\end{tikzcd}
$$

$$
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\theta' \arrow[dr] & & \\
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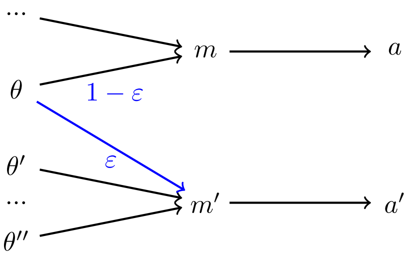
Consider any other message m' and an alternative strategy $\widehat{\sigma}_S$:



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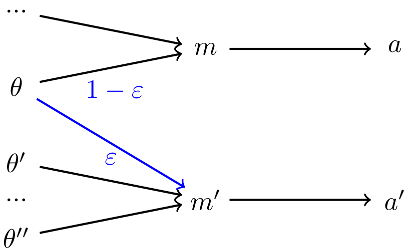
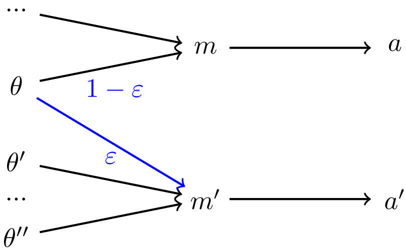
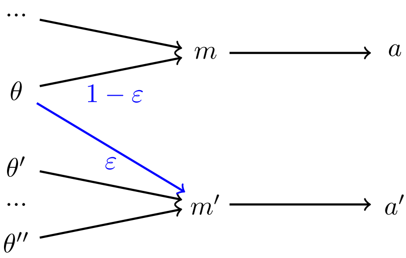


Since a and a' are strict best responses, R-IC is satisfied for small ε.



The Sender's payoff change under such a revision is

$$\varepsilon[u_S(\theta, \sigma_R^*(m')) - u_S(\theta, \sigma_R^*(m))]$$



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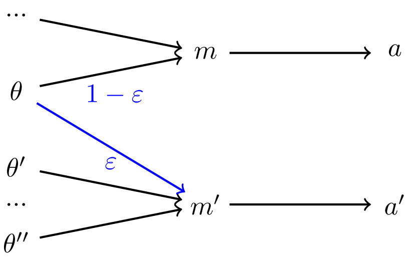

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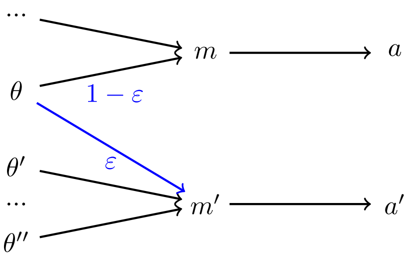

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$\Longrightarrow$ (σ_S^*, σ_R^*) is a cheap-talk equilibrium.

Proof of “Only If” and Theorem 2

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Recall:

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- Theorem 2: commitment has no value $\Rightarrow$ cheap-talk Sender does not value randomization

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- Theorem 2: commitment has no value $\Rightarrow$ cheap-talk Sender does not value randomization

Both Theorem 2 and the “Only If” direction of Theorem 1 follow from:

Proposition. If (σ_S, σ_R) is a cheap-talk equilibrium that yields the persuasion payoff, then we can construct a pure-strategy cheap-talk equilibrium $(\widehat{\sigma}_S, \widehat{\sigma}_R)$ that yields the persuasion payoff.

The proposition follows via two steps (both hold generically):

Lemma 1. If (σ_S, σ_R) is an R-IC profile and yields the persuasion payoff, then σ_R must be a pure strategy.

Lemma 2. If (σ_S, σ_R) is a cheap-talk equilibrium that yields the persuasion payoff and σ_R is a pure strategy, then there exists a partitional cheap-talk equilibrium that yields the persuasion payoff.

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- Receiver randomizes so must be indifferent
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Claim: generically, such indifference cannot arise if Sender optimizes

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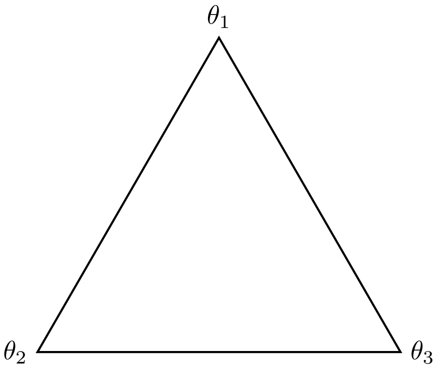

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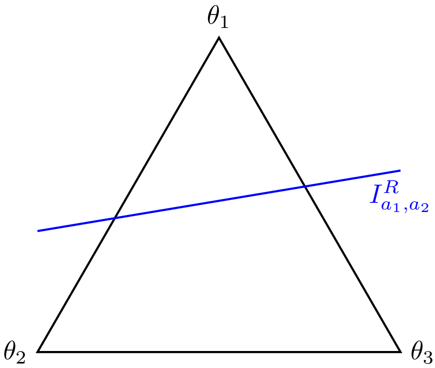

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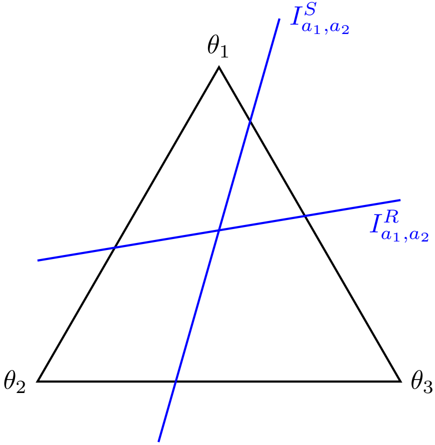

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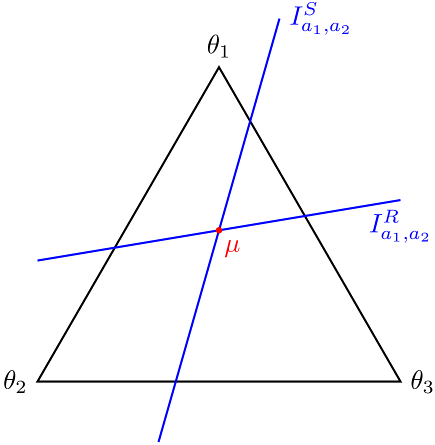

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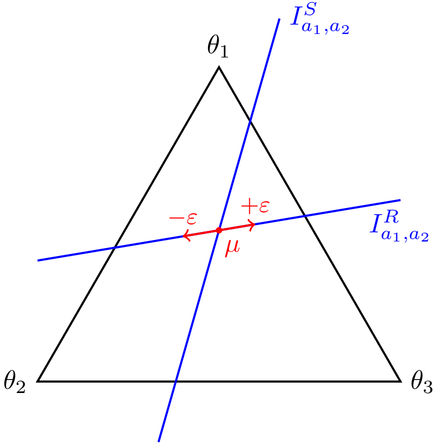

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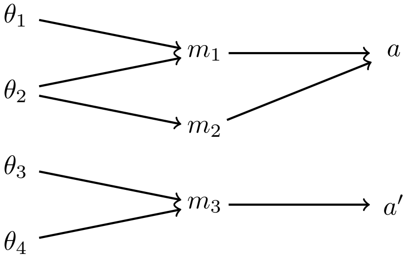
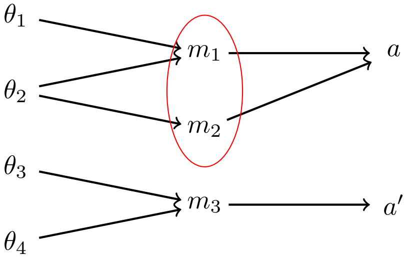
If $\sigma_S(\cdot|\theta)$ plays both m and m', these two messages must induce the same action.


θ1
θ2
θ3
θ4
m1
m2
m3
a
a′



Let $M = A$, $\widehat{\sigma}_S(\theta) = \sigma_R(\sigma_S(\theta))$, $\widehat{\sigma}_R(a) = a$.

$(\widehat{\sigma}_S, \widehat{\sigma}_R)$ is also a cheap-talk equilibrium and also yields persuasion payoff.

How often?

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We say preferences are felicitous if for each non-empty Θ_i

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λ(Commitment has no value) $\geq$ λ(felicity)

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$$\lambda\left\{u_R \in [0,1]^{\Theta_1 \times A} | \sum_{\theta \in \Theta_1} \mu_0(\theta) u_R(\theta, a_1) \geq \sum_{\theta \in \Theta_1} \mu_0(\theta) u_R(\theta, a_2)\right\} = \frac{1}{2}$$

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If one of Θ_i is empty,

$$\lambda(\text{felicity}) = \frac{1}{2}$$

$$\lambda(\text{commitment has no value}) \geq \lambda(\text{felicity}) \geq \frac{1}{4}$$

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As $|\Theta| \to \infty$, both "$\geq$" converge to "$=$"

Large state space

λ(commitment has no value) $\rightarrow$ λ(felicity)

Large state space

λ(commitment has no value) $\to$ λ(felicity)

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As $|\Theta| \rightarrow \infty$, λ(joint-inclusivity) $\rightarrow 1$.

Large state space

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- For each θ and a, the share of preferences such that a is ideal for Sender and Receiver in state θ is $\frac{1}{4}$.

Large state space

λ(commitment has no value) $\to \lambda$(felicity)

An environment is jointly-inclusive if for every a, there exists θ such that a is the ideal action for both Sender and Receiver in state θ.

As $|\Theta| \to \infty$, λ(joint-inclusivity) $\to 1$.

- For each θ and a, the share of preferences such that a is ideal for Sender and Receiver in state θ is $\frac{1}{4}$.
- For each a, the share of preferences such that a is never an ideal action for both players in any state is

Large state space

λ(commitment has no value) $\to \lambda$(felicity)

An environment is jointly-inclusive if for every a, there exists θ such that a is the ideal action for both Sender and Receiver in state θ.

As $|\Theta| \to \infty$, λ(joint-inclusivity) $\to 1$.

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$$\left(1 - \frac{1}{4}\right)^{|\Theta|} \to 0.$$

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Lemma. If commitment has no value in a jointly-inclusive environment, then the environment is felicitous.

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$\Rightarrow \lambda(\text{felicity}) \to \frac{1}{4}$

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Connection between how often commitment has no value and the cardinality of the action set.

- λ(commitment has no value) $\geq \frac{1}{|A|^{|A|}}$
- As $|\Theta| \to \infty$, λ(commitment has no value) $\to \frac{1}{|A|^{|A|}}$

Literature

Under Transparent Motives:

- Sender obtains ideal payoff through cheap talk or Sender values commitment (Lipnowski-Ravid '20)
- Sender obtains ideal payoff at prior or Sender values randomization (Best-Quigley '24)
- Cheap talk = Persuasion $\Leftrightarrow$ Mediation = Persuasion (Corrao-Dai '24)
- Verifiable messages = Persuasion $\Rightarrow$ Randomization has no value (Titova-Zhang '25)

Commitment and Randomization by Receiver in Disclosure Games:

- Glazer-Rubinstein '06; Sher '11; Hart-Kremer-Perry '17